

Abhishek Dangwal

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Skills

Languages:	Python, JavaScript, Java, Scala, C++, SQL, Bash, R
ML & AI Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, PEFT, LoRA, QLoRA, TRL, SentenceTransformers, LangChain, LlamaIndex, CrewAI, Google ADK, OpenAI SDK, XGBoost, LightGBM, CatBoost, Optuna, MLflow, Weights & Biases, NumPy, Pandas, SciPy, Statsmodels, OpenCV, ONNX Runtime, vLLM, Unsloth, bitsandbytes, FAISS, Qdrant, BM25, Ragas, DeepEval, Promptfoo
Data Eng & MLOps:	Apache Spark, Databricks, Hadoop, Airflow, Docker, Kubernetes, Git, GitHub Actions, Terraform, AWS (S3, EC2, SageMaker, Glue, Redshift), GCP (BigQuery, Vertex AI, Cloud Functions), Azure ML, Snowflake, Qdrant, Faiss, Kafka
Analytics & Visualization:	Tableau, Power BI, Looker Studio, Superset, Excel, Jupyter, Plotly, Matplotlib, Seaborn, Streamlit, Dash
Core Data-Science Skills:	Machine Learning, Deep Learning, Generative AI (LLMs, SLMs, VLMs), NLP, Computer Vision, Multimodal AI, Retrieval-Augmented Generation (RAG), Agentic AI Systems, Multi-Agent Orchestration, Memory & State Management, Tool/Function Calling, Prompt Engineering, Structured Prompting, LLM Evaluation (Ragas, DeepEval), Model Fine-tuning (LoRA, QLoRA, PEFT), Embeddings & Semantic Search, Feature Engineering, Predictive Modeling, Time-Series Forecasting, Anomaly Detection, Statistical Analysis, A/B Testing & Experimentation, Causal Inference, Model Interpretability (SHAP, LIME), Explainable AI (XAI), Data Modeling, Data Warehousing, ETL/ELT Pipelines, Data Governance, Data Quality, Real-time Data Processing, Decision Systems, Optimization & Cost-Aware Model Routing

Work Experience

Freecharge, India Feb 2026 – Present

Lead Data Scientist – AI/ML

- Led the design and deployment of **BSA (Bank Statement Analyser)**, an end-to-end **agentic loan decisioning system** processing PDFs and JSONs using **OCR + LLM/VLM pipelines**, replacing third-party APIs (Perfios) and reducing per-call cost from INR 70 to under INR 10 (**85% cost reduction**); leveraged **Google ADK, AWS Textract, and custom agent orchestration frameworks**.
- Architected a **Financial Compliance Underwriting (FCU) engine** integrated with BSA, detecting **50+ suspicious behavioral and financial risk patterns** using **hybrid ML + LLM reasoning + rule-based systems**, with explainable outputs and audit trails for underwriting workflows.
- Built advanced **PDF forensic analysis pipelines** combining **computer vision, signal processing, and transformer-based anomaly detection** to detect tampering (metadata inconsistencies, font anomalies, layout shifts, recompression artifacts), improving fraud detection precision in financial documents.
- Fine-tuned **Vision-Language Models (VLMs)** (CLIP-style, encoder-decoder multimodal transformers) for tampering detection across **Indian financial and identity documents** (Aadhaar, PAN, bank statements) using **LoRA, QLoRA, PEFT, contrastive learning, and layout-aware embeddings**.
- Designed and deployed a **secure agentic financial intelligence system** for Axis Bank using **on-prem SLMs (Small Language Models), RAG + memory layers, and tool-augmented agents**, enabling:
 - Personalized expense tracking and semantic transaction categorization
 - Temporal pattern mining and behavioral anomaly detection
 - Predictive budgeting, savings optimization, and financial goal planning
 - Upcoming bill forecasting using time-series + LLM hybrid reasoning
 - Risk scoring and fraud alerts via ensemble ML + LLM validation
- Built an **AI-powered conversational banking agent** integrated within the Axis mobile app using **multi-modal agents (text + vision + speech)**, enabling:
 - Beneficiary creation via cheque/image ingestion (OCR + KYC validation pipelines)
 - Secure payments via voice/chat with **intent classification + tool calling**
 - Context-aware account insights, summaries, and recommendations
 - Autonomous workflows using **function calling, tool APIs, and stateful agents**
- Architected **multi-agent systems** (planner–executor–critic/validator) with **memory (short-term + long-term), state management, and deterministic execution**, reducing hallucination and improving reliability in high-stakes financial workflows.
- Built production-grade **RAG pipelines** using **Qdrant, FAISS, hybrid BM25 + dense retrieval, cross-encoder rerankers**, query rewriting, and structured output parsers for financial document intelligence.
- Leveraged **transformer architectures** (encoder, decoder, encoder-decoder) across use-cases including classification,

retrieval, multimodal reasoning, and sequence generation.

- Designed scalable AI microservices using **FastAPI, Docker, Kubernetes**, with distributed async pipelines for large-scale document ingestion, OCR, chunking, embedding generation, and vector indexing.
- Implemented **LLMOps and MLOps pipelines**: prompt versioning, evaluation frameworks (**Ragas, DeepEval**), hallucination tracking, model registry, CI/CD (GitHub Actions), and observability (Prometheus, OpenTelemetry).
- Optimised inference cost and latency via **quantisation, batching, caching, distillation, and SLM vs LLM routing strategies**, balancing performance, cost, and security.
- Applied **advanced prompting techniques** (structured prompting, chain-of-thought, self-reflection, LLM-as-a-judge) to improve reasoning accuracy and output reliability.
- Integrated **vector databases, relational systems, and streaming pipelines** (PostgreSQL, Elasticsearch, event queues) to support real-time AI-driven financial applications.
- Collaborated with banking stakeholders, underwriting teams, and product leaders to translate regulatory, compliance, and financial requirements into **production-grade AI systems**.

Forage, New Jersey, USA Dec 2024 – Jan 2026

AI Developer

- Fine-tuned **multimodal LLMs/VLMs and SLMs** using PyTorch, Hugging Face, LoRA/QLoRA, PEFT, and Vertex AI; optimized serving with quantization, ONNX, batching, and low-latency inference patterns.
- Architected **agentic AI systems** using CrewAI, LangChain, LlamaIndex, and OpenAI Assistants with planner-executor-verifier flows, tool calling, memory, and stateful orchestration.
- Built production-grade **RAG pipelines** with Qdrant, FAISS, hybrid BM25+dense retrieval, query rewriting, rerankers, metadata filters, contextual chunking, and structured outputs.
- Designed scalable **LLM microservices** with FastAPI, Docker, Kubernetes (GKE), Ray, and cloud-native queues for async ingestion, document processing, embedding, and autoscaled inference.
- Owned **LLMOps/MLOps** end to end: MLflow/W&B experiments, prompt/version control, eval datasets, CI/CD, guardrails, rollback strategies, and observability using Prometheus, Grafana, and OpenTelemetry.
- Applied **trending GenAI reliability techniques** including LLM-as-a-Judge, self-reflection, synthetic test generation, prompt regression testing, schema validation, and hallucination/latency/retrieval-quality evaluation.
- Improved agent accuracy through **advanced retrieval and reasoning patterns** such as adaptive context windows, cross-encoder reranking, multi-agent role separation, and metadata-aware memory retrieval.
- Built supporting **classical ML and analytics workflows** using XGBoost, LightGBM, logistic regression, pandas, NumPy, SQL, and BI tools for prediction, experimentation, and stakeholder reporting.

Clootrack, Bangalore

Oct 2022 – Dec 2024

Data Analyst

- Executed exploratory data analysis and statistical modeling with Python (pandas, NumPy, statsmodels), SQL, and **dbt (data build tool)** for data transformation and feature engineering across multi-source datasets.
- Built modular NLP pipelines using NLTK, transformers, and Sentence-BERT for sentiment/topic extraction across customer-journey data; implemented **dbt models** to clean, join, and version curated datasets for analytics.
- Created interactive dashboards in Tableau and Power BI, blending data from BigQuery, Google Sheets, Snowflake, and REST/GraphQL APIs; managed data lineage with **dbt sources, tests, and documentation**.
- Automated data ingestion and ELT workflows using Selenium, Requests, Airflow, and **dbt Cloud**; enforced data quality through dbt tests (unique, not null, relationships) and Git-based CI/CD.
- Analyzed marketing campaign performance metrics (ROAS, CTR, CPC, CVR) using regression, correlation analysis, and **SQL/dbt incremental models** to optimize ad spend and maximize return.

Benchmark, Gurgaon

Oct 2021 – Oct 2022

Technical Analyst

- Investigated and resolved email-service deliverability issues using SQL, Python, and HTML/CSS QA to improve campaign stability.
- Automated audience segmentation, workflow orchestration, and A/B testing with HubSpot, Marketo APIs, and Zapier for personalized outreach.
- Built data-quality dashboards with Excel Power Query and Looker Studio to provide stakeholders with up-to-date campaign metrics.

My Projects

- **Language Detection — XLM-RoBERTa Pipeline** [GitHub]
 - Fine-tuned **XLM-RoBERTa (transformer-based multilingual model)** with Optuna hyperparameter tuning, achieving 93% macro-F1 across 100+ languages.
 - Built scalable **NLP inference pipelines** using joblib, memory mapping, and batch processing for multi-GB datasets.

- Deployed production-ready **ML microservices** using FastAPI, Docker, and Cloud Run with observability (Prometheus, Grafana).
- **Punctuation Restoration & Text Enhancement (NLP Pipeline)** [GitHub]
 - Built a **sequence-to-sequence NLP system** using T5 transformers, spaCy alignment, and rule-based post-processing for ASR text normalization.
 - Optimised throughput using **parallel processing (torch.multiprocessing)** achieving ~1M sentences/hour.
 - Packaged as a modular Python library with **config-driven pipelines (YAML), testing (pytest), and reusable components**.
- **Project Contextualizer (Agentic Prompt Optimization Engine)** [GitHub]
 - Developed an **agentic prompt engineering system** for **context compression, structured prompting, and prompt compilation** for LLM workflows.
 - Implemented **semantic segmentation, summarisation, and transformation pipelines** to optimise token usage while preserving reasoning quality.
 - Built evaluation layer for **prompt efficiency metrics** (token reduction, structure quality, reasoning retention) for production GenAI systems.
- **Knowledge Graph-based Fact Checker (Explainable AI System)** [GitHub]
 - Designed a **fact verification system** combining **knowledge graphs, NLP, and graph-based reasoning** for claim validation.
 - Implemented **entity extraction, relation mapping, and graph traversal algorithms** for structured evidence retrieval.
 - Generated **explainable AI outputs (XAI)** with traceable reasoning paths, improving trust over black-box LLM responses.
- **RAG PDF Dashboard (Document Intelligence Platform)** [GitHub]
 - Built an end-to-end **Retrieval-Augmented Generation (RAG)** system using **vector databases (FAISS/Qdrant), embeddings, and hybrid search**.
 - Implemented document pipelines (chunking, semantic indexing, retrieval, reranking) for **context-aware LLM responses**.
 - Developed an interactive dashboard enabling **document Q&A, semantic search, and knowledge extraction** across large corpora.
- **Agentic Classifier Studio (Human-in-the-Loop AI System)** [GitHub]
 - Built a **multi-agent classification framework** with **LLMs, memory, and tool usage** enabling interactive model training and validation.
 - Implemented **Chain-of-Thought (CoT), feedback loops, and human-in-the-loop learning** for improved classification accuracy.
 - Designed modular pipelines for **dataset ingestion, prompt versioning, agent state management, and reusable inference workflows**.
- **Sales Data AI Agent (Autonomous Analytics System)** [GitHub]
 - Developed an **agentic analytics system** leveraging **LLMs + Python tool execution** for automated EDA and business insights generation.
 - Enabled natural language querying with **data pipelines (pandas, NumPy)** for KPI tracking, trend analysis, and anomaly detection.
 - Implemented **multi-step reasoning agents** for decision support, recommendations, and automated reporting workflows.

Education

Uttarakhand Technical University
Bachelors of Technology

Subharti University Meerut
Diploma in Engineering

Awards and Certificates

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| • British Airways Data Science Job Simulation | • Foundations: Data, Data, Everywhere |
| • MySQL | • Microsoft Excel |
| • Amazon EC2 Observability | • Prepare Data for Exploration |
| • Google Analytics | • KPMG Data Analytics Consulting Virtual Internship |
| • Ask Questions to Make Data-Driven Decisions | |